SP-DETR: Saliency-Guided Sparse Encoding and Group Policy Optimization for SAR Object Detection

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***Abstract*—Target responses in synthetic aperture radar (SAR) images are often locally prominent and spatially discontinuous and can be easily mixed with background clutter and false high-response regions. Meanwhile, weak boundaries and incomplete responses introduce ambiguity in localizing target extents. To address the limited adaptability of generic Detection Transformer (DETR) detectors to these SAR object detection challenges, this article proposes SP-DETR, an end-to-end SAR object detector built upon RT-DETR. To handle the mixture of target-related cues and background tokens in shallow high-resolution features, a saliency-guided sparse encoder (SGSE) is designed. It estimates target relevance at different spatial locations using multiscale saliency maps, dynamically selects high-value tokens for efficient feature interaction, and reweights fused features with saliency information to enhance target-related responses while suppressing background interference. To improve localization stability under weak boundaries and degraded target appearances, a Group Policy Optimization Decoder (GPOD) is further introduced. It models predicted bounding boxes as Gaussian distributions and optimizes candidate box quality through multigroup sampling and a global localization reward constraint, thereby improving localization robustness in complex scenes. Experiments on SAR-AIRCraft-1.0 and SARDet-100K demonstrate that SP-DETR achieves the best overall performance among the compared methods. Ablation studies, efficiency comparisons, and visualization results further verify the effectiveness of the proposed modules.**

***Index Terms*—Detection Transformer (DETR), reinforcement learning, sparse attention, synthetic aperture radar (SAR) object detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) offers day-and-night, all-weather, long-range, and high-resolution observation capabilities and has, therefore, become an important tool for Earth observation [1]. With the continued maturation of SAR imaging techniques and the deployment of airborne and satellite SAR systems, high-quality SAR imagery has become increasingly available. These data provide substantial support for the intelligent interpretation of SAR imagery, including object detection, target recognition, and SAR-to-optical image translation [2], [3], [4], [5], [6], [7], [8], [9].

Received 25 June 2026; revised 14 August 2026; accepted 25 August 2026. Date of publication 2 September 2026; date of current version 11 September 2026. This work was supported by the National Natural Science Foundation of China under Grant 62425113. *(Corresponding author: Xueru Bai.)*

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Digital Object Identifier 10.1109/TGRS.2026.3729565

As an important component of SAR image interpretation, SAR object detection aims to identify the locations and categories of targets of interest [10]. It plays an important role in applications such as battlefield reconnaissance and situational awareness and remains an active research topic in the SAR community. A recent survey reviews 50 years of SAR object detection and recognition, covering representative methods, datasets, evaluation metrics, challenges, and future directions for benchmarks and open-source ecosystems [11]. Traditional SAR target detection and recognition methods are primarily based on the three-stage processing pipeline developed at Lincoln Laboratory [12], [13]. Although these methods can achieve satisfactory performance in simple scenes, they are less suited to complex scenarios and the large volumes of SAR data currently available. With rapid advances in deep learning [14], [15], [16], numerous studies have applied deep neural networks to SAR object detection. These methods have achieved substantial improvements in both speed and accuracy over the traditional three-stage pipeline and have become the dominant approach in this field.

Most deep learning-based SAR object detectors adopt CNN-based one- or two-stage architectures [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]. Although convolution is effective at local feature modeling, these detectors have limited ability to capture long-range dependencies. In addition, their one-to-many label assignment usually produces redundant predictions and makes them reliant on postprocessing operations, such as nonmaximum suppression (NMS) [27]. In contrast, the Detection Transformer (DETR) framework [28], [29], [30] models object detection as a set prediction problem. Its one-to-one label assignment through Hungarian matching removes the need for postprocessing operations such as NMS, thereby enabling end-to-end detection. Although early DETR variants suffered from slow convergence and high inference latency, subsequent advances have led to RT-DETR [31], which achieves fast convergence and competitive speed and accuracy.

Nevertheless, directly applying general-purpose DETR detectors designed for natural images to SAR imagery remains challenging. Existing saliency-driven sparse DETRs primarily use saliency to rank and filter redundant encoder tokens or queries for efficient computation [32], [33]. SAR detection poses a different requirement: sparse target-related scattering cues must be preserved from shallow high-resolution features,

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while numerous clutter-dominated responses are prevented from interfering with feature interaction. In addition, weak or incomplete target boundaries make the spatial extent of a target ambiguous. These characteristics expose two limitations of existing DETR encoders and decoders when they are applied to SAR imagery.

First, target-related responses in SAR imagery are often spatially sparse and locally discontinuous. Localized scattering responses, edge structures, and shadow variations are more likely to be preserved in shallow high-resolution features, but these features also contain speckle noise, background scattering, sidelobe responses, and pseudotarget highlights. Encoding only deeply downsampled features can weaken local target details [34], [35], whereas densely updating all shallow tokens is computationally expensive and allows clutter-dominated tokens to propagate irrelevant responses. Moreover, response strength at a single shallow level is insufficient for reliable token selection because both targets and clutter may produce locally strong activations. Deep features provide more stable target semantics but lack precise spatial details. This motivates a hierarchical mechanism in which deep semantic saliency guides shallow token selection and shallow local–global cues progressively refine the saliency at each resolution.

Recent optical remote-sensing salient object detection research has also demonstrated the value of multiscale saliency modeling. MSDEN models local center–surround contrast through multiscale difference excitation, refines boundary cues with high-level semantic guidance, and performs compact cross-scale feature fusion [36]. These designs highlight the importance of coordinating semantic information with fine spatial details in remote-sensing imagery.

Second, SAR target boundaries may be weak, incomplete, or displaced by bright scattering centers, shadows, and surrounding clutter. Consequently, several box extents can be locally plausible for the same target. Mainstream DETR decoders regress one deterministic box for each query, making localization sensitive to a dominant but incomplete local response [37], [38]. Multiroute training methods such as Mr. DETR improve decoder supervision through parallel one-to-one and one-to-many assignment tasks [29], but their deterministic routes do not explicitly explore alternative box extents for a fixed query-target correspondence. This motivates stochastic candidate exploration under a consistency bipartite matching (CBM) strategy so that localization alternatives can be evaluated without changing the matching target of each query.

Based on the first insight, we design a saliency-guided sparse encoder (SGSE) that recursively propagates aligned saliency from deep semantic features to shallow high-resolution features and refines it at each level using local and global cues. Unlike using saliency only as a token-filtering score, SGSE applies the resulting hierarchical maps at two stages. They first select target-relevant shallow tokens for sparse feature interaction, reducing both computation and the propagation of clutter responses, and then modulate the fused multiscale features to attenuate residual background interference.

Based on the second insight, we introduce a Group Policy Optimization Decoder (GPOD). GPOD models the box predicted by each query as a Gaussian distribution and draws multiple complete prediction sets from the same joint policy while retaining the Hungarian assignment computed from the predicted means. For each sampled set, the image-level reward averages the normalized GIoU of all matched targets and applies the resulting advantage to their joint localization policy, rather than assigning an independent reward to each sampled box. This image-level credit assignment complements the instancewise supervised loss by jointly evaluating the localization quality of all matched predictions. Thus, the multiple samples explore alternative extents under a fixed query–target correspondence, rather than serving as separate deterministic decoder routes or assignment tasks. Only the predicted means are used at inference.

In summary, the main contributions of this work are given as follows.

1) We propose SP-DETR, an end-to-end SAR object detection framework based on RT-DETR. It optimizes the encoder and decoder for sparse local target responses, clutter interference, and localization ambiguity in SAR imagery, improving detection accuracy and localization robustness in complex SAR scenes.
2) We design an SGSE that propagates deep semantic saliency to shallow high-resolution features and progressively refines it with local–global cues. The hierarchical saliency maps guide both sparse shallow-token interaction and soft attention mask modulation, preserving target details while reducing computation and clutter interference.
3) We propose a GPOD that samples multiple box sets from a shared Gaussian localization policy under fixed query-target assignments and optimizes them using an image-level reward. This stochastic candidate exploration improves box optimization when weak SAR boundaries admit multiple plausible target extents.
4) Experiments on SAR-AIRCraft-1.0 [4] and SARDet-100K [6] show that SP-DETR achieves the best overall $AP_{50:95}$ among the compared methods. Ablation studies, efficiency comparisons, and visualization analyses further assess the effects of SGSE and GPOD on feature selection, localization optimization, and computational cost.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the proposed SP-DETR object detection model. Section IV reports the experimental results and analyses. Finally, Section V concludes this article and outlines future work.

II. Related Work

A. SAR Object Detection

Benefiting from the powerful representation capability of deep neural networks, deep learning-based SAR object detection methods have achieved clear advantages over traditional approaches in both detection speed and accuracy. Consequently, they have garnered extensive attention from researchers in recent years. To cope with complex backgrounds and clutter interference in SAR object detection, CenterNet++

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[20] extracts features enriched with stronger semantic information via feature adjustment and feature pyramid fusion modules, subsequently mitigating the impact of clutter on ship detection through foreground–background balancing. Similarly, RDB-DINO [39] alleviates the confusion between ship targets and the background regions through a complete contrastive denoising training technique. To tackle the ubiquitous noise interference inherent in SAR images, DenoDet [40] implements dynamic denoising across multiple subspaces via a dynamic frequency-domain attention mechanism. Meanwhile, MD-DETR [41] effectively suppresses the adverse effects of speckle noise through a multilevel denoising strategy operating at the image, feature, and query levels. More recently, MaDiNet [42] employs a Gamma diffusion model to generate target bounding boxes and introduces MambaSAR to capture spatial structural information for distinguishing targets from complex backgrounds. Furthermore, to bridge the substantial gap between SAR and optical imagery, which limits the direct transfer of general-purpose detectors, CRTransSar [43] and SFS-CNet [44] design novel backbone architectures to enhance feature extraction for SAR imagery. MSFA [6] mitigates the discrepancy between SAR and optical modalities through multilevel pretraining on both optical and remote sensing images. In addition, leveraging optical remote sensing images to assist SAR object detection has increasingly gained traction. For instance, CoLD [45] employs knowledge distillation, using an optical detection model to supervise the training of the corresponding SAR model. Moreover, a progressively transferred unsupervised domain adaptation framework [46] enables SAR ship detection through knowledge transfer from the optical domain to the SAR domain. SM3Det [47] incorporates a sparse mixture-of-experts module and a dynamic learning rate adjustment strategy to acquire cross-modal transferable knowledge among optical, infrared, and SAR images, thereby achieving performance improvements across datasets from all corresponding domains.

However, despite the progress of existing SAR object detection methods, the selective utilization of local target cues in shallow features under background interference and robust box optimization under ambiguous target extents remain insufficiently explored.

B. Detection Transformer

By leveraging the Transformer architecture and the Hungarian algorithm, DETR [28] reformulates the object detection problem into a set prediction task. This paradigm removes the reliance on handcrafted components, such as anchor boxes and NMS, thereby enabling genuine end-to-end object detection. Building on this basis, a variety of methods have been developed to overcome its inherent limitations. To tackle the issues of slow convergence and inadequate capability in detecting small objects, Deformable DETR [48] designs deformable attention, which accelerates model convergence while enhancing small object detection. DAB-DETR [49] accelerates convergence by parameterizing object queries with learnable dynamic anchors, while DINO [50] further improves detection performance through the joint use of denoising-based contrastive supervision and a hybrid query selection scheme.

Furthermore, the one-to-one label assignment inherent to the Hungarian algorithm often leads to a scarcity of positive samples during training. To alleviate this, Co-DETR [51] increases the proportion of positive samples by constructing auxiliary branches with one-to-many label assignment and employing collaborative hybrid training. Mr. DETR [29] treats the DETR model as a multitask framework, optimizing the training process by executing one-to-one and one-to-many prediction tasks in parallel. Unlike Mr. DETR's deterministic multitask routes, GPOD samples multiple box sets from a shared Gaussian localization policy while retaining the Hungarian assignment computed from the predicted means and optimizes their image-level localization reward. Addressing the lack of supervision information during training, DEIM [34] realizes dense one-to-one assignment with the help of mosaic and mixup augmentation schemes.

Finally, to overcome the suboptimal inference speed of DETR models, RT-DETR [31] accelerates inference by devising a lightweight hybrid encoder and a low-uncertainty query selection strategy. Meanwhile, Sparse DETR [32], Focus DETR [52], and Salience DETR [33] improve inference efficiency by incorporating saliency supervision signals to decrease the encoder token load. Sparse DETR selects decoder-relevant tokens using cross-attention-derived pseudolabels, whereas Salience DETR reduces scale bias through scale-independent supervision and hierarchical filtering ratios. SGSE instead propagates deep semantic saliency to refine shallow saliency and applies level-specific encoding: selected shallow tokens undergo sparse deformable interaction, whereas deep features retain full self-attention. The saliency maps also suppress clutter after multiscale fusion.

C. Uncertainty-Aware Localization

To overcome the limitations of traditional deterministic regression in handling ambiguous target boundaries, uncertainty modeling has increasingly become an active area of research. Early works, such as KL-Loss, are primarily grounded in the assumption of parameterized Gaussian distributions [53]. These methods adaptively learn the localization variance of object boxes by minimizing the KL divergence from the estimated distribution to the ground-truth Dirac delta distribution. This learned variance is subsequently employed for weighted voting during the NMS stage. Similarly, Gaussian YOLOv3 [54] models bounding box coordinates as Gaussian distributions, enhancing detection robustness by minimizing the negative log-likelihood. Recognizing that the rigid Gaussian assumption often struggles to accurately fit the complex and diverse distributions of real-world boundaries, generalized focal loss [37], [38] introduced discrete probability distribution modeling. By learning probability curves of arbitrary shapes to capture boundary ambiguity, GFL utilizes distribution statistics to guide the quality assessment of bounding boxes. More recently, D-FINE [55] adapted this fine-grained distribution modeling paradigm into the DETR architecture, significantly boosting the localization accuracy of the end-to-end detector through iterative refinement of distribution residuals within the decoder.

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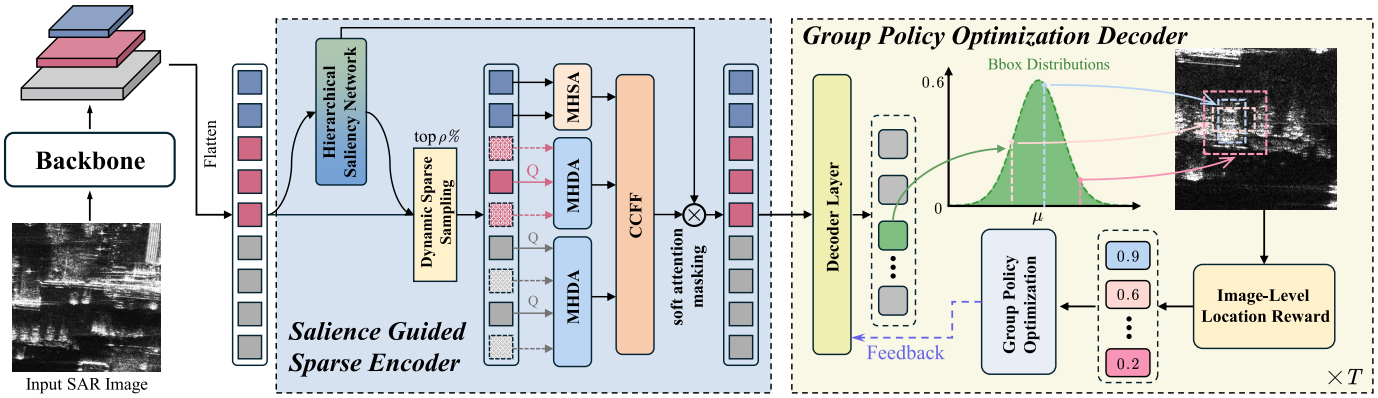


Fig. 1. Overview of the SP-DETR architecture. In the framework, multiscale features generated by the backbone are first passed to the SGSE. Within the SGSE, an HSN predicts saliency scores indicating the presence of targets. Utilizing these scores as spatial priors, the module samples shallow features and executes sparse attention interaction. Subsequently, the features filtered and enhanced by the SGSE are transformed into queries and fed into the GPOD. Ultimately, through the synergistic optimization of supervised learning and reinforcement learning, the GPOD progressively decodes the categories and precise locations of the targets.

Existing uncertainty-aware detectors mainly learn or refine per-box localization distributions through supervised objectives [37], [38], [53], [54], [55]. GPOD uses the predicted distributions differently, treating them as a stochastic localization policy from which multiple complete box sets are sampled. A CBM strategy computed from the predicted means preserves query–target correspondence across sampling groups, while an image-level localization reward evaluates each sampled set. This formulation explores alternative target extents when weak or incomplete SAR boundaries make deterministic regression sensitive to local scattering responses and surrounding clutter.

III. Method

A. Overall Structure

An overview of the SP-DETR architecture is presented in Fig. 1. Following an end-to-end detection paradigm, it comprises three primary modules: a feature extraction backbone, an SGSE, and a GPOD. Initially, the backbone extracts features at multiple scales from the input SAR image. Subsequently, the resulting feature representations are passed to the hierarchical saliency network (HSN) in the SGSE to generate saliency scores for each feature level. After dynamically filtering the key features corresponding to the top $\rho\%$ of saliency scores at each scale, sparse interaction and feature fusion are executed. Thereafter, the fused features outputted by the CCFF module are modulated using the corresponding saliency maps as soft attention masks. This modulation substantially improves the model's ability to capture local target details. Finally, the modulated features are transformed into queries and injected into the GPOD to progressively decode the categories and precise locations of the targets. Furthermore, to effectively guide the model training, SP-DETR employs a hybrid loss function comprising a saliency loss and a decoder loss. Specifically, the saliency loss is applied to the HSN to supervise the generation of saliency scores, while the decoder loss is utilized to supervise target localization and classification.

The specific architectures and implementation details of the SGSE and GPOD are detailed in the following.

B. Saliency-Guided Sparse Encoder

SGSE first predicts hierarchical saliency maps through cross-level saliency guidance and then uses them for sparse token interaction and soft attention mask modulation. At shallow levels, only selected high-saliency tokens undergo attention-based updating, while unselected tokens are preserved through identity mapping.

1) Saliency Prediction: The SGSE takes as input a set of flattened multilevel feature maps, denoted by $\mathcal{V} = \{\boldsymbol{f}^l\}_{l=1}^{L}$, where L is the number of feature levels. As shown in Fig. 2, HSN applies a shared saliency predictor recursively in a top–down order. At the deepest level, s^L is predicted directly from $\boldsymbol{f}^L$. For each shallower level $l = L-1, \ldots, 1$, the aligned saliency score s^{l+1} from the next deeper level provides a coarse semantic prior for predicting s^l from $\boldsymbol{f}^l$. This cross-level saliency guidance transfers target-level semantics to higher-resolution features while allowing each level to retain its scale-specific spatial details.

The saliency predictor is detailed on the right-hand side of Fig. 2. It employs a dual-path aggregation structure, enabling the features to capture both global and local characteristics. First, the feature $\boldsymbol{f}^l$ is modulated using the saliency score s^{l+1}

$$\tilde{\boldsymbol{f}}^l = \boldsymbol{f}^l + \beta^l \cdot \left(\Psi\left(s^{l+1}\right) \cdot \boldsymbol{f}^l\right) \tag{1}$$

where β^l is a learnable coefficient that adjusts the influence of s^{l+1} on the features and $\Psi(\cdot)$ denotes the spatial alignment operator employed to achieve spatial alignment of saliency scores across different levels, formulated as

$$\Psi\left(s^{l+1}\right) = \mathrm{vec}\left(\mathcal{U}\left(\mathcal{R}\left(s^{l+1}\right)\right)\right) \tag{2}$$

where $\mathcal{R}(\cdot)$ represents the dimension transformation operator, converting saliency scores into saliency maps; $\mathcal{U}(\cdot)$ represents the upsampling operation for resizing them to the spatial resolution of the lth level; and $\mathrm{vec}(\cdot)$ signifies vectorization, transforming the saliency maps back into saliency scores. At

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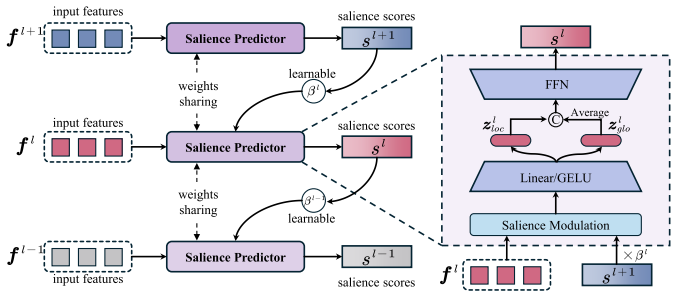


Fig. 2. Architecture of the HSN. The deepest saliency score is predicted directly from the deepest feature. At each shallower level, the aligned saliency score from the next deeper level modulates the current feature through a learnable residual connection, and the local–global predictor refines this semantic prior at the current resolution. The resulting saliency maps guide dynamic sparse sampling and soft attention mask modulation.

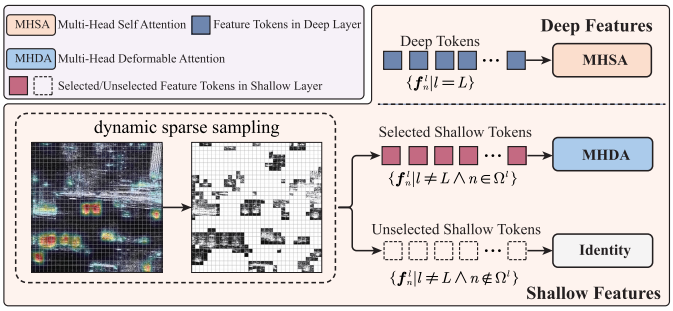


Fig. 3. Overview of the sparse interaction strategy in SGSE. For shallow features, informative tokens are dynamically selected for multihead deformable attention to preserve local target details, while unselected tokens are retained via identity mapping. For deep features, multihead self-attention is employed to strengthen high-level semantic representation.

spatial position n, (1) can be written as $\tilde{\boldsymbol{f}}_n^l = [1+\beta^l \Psi(s^{l+1})_n]\boldsymbol{f}_n^l$. The aligned deep-level saliency, therefore, acts as a coarse semantic prior whose spatial values control the magnitude of the residual modulation. Shallow responses with stronger high-level semantic support receive a larger semantic-conditioned adjustment, whereas isolated responses with weak support remain closer to their original representations. Because the residual connection does not hard-filter low-saliency positions, weak or discontinuous scattering details are retained for subsequent refinement. The local–global predictor then reestimates saliency at the current resolution, allowing the spatial structures in $\boldsymbol{f}^l$ to refine rather than simply copy the coarse prior.

Subsequently, the dual-path aggregation structure projects the modulated feature $\tilde{\boldsymbol{f}}^l$ into two complementary components, namely, the local features $\boldsymbol{z}_{loc}^l$ and the global features $\boldsymbol{z}_{glo}^l$. In this way, local details and broader contextual information can be modeled jointly, which is formulated as follows:

$$\left[\boldsymbol{z}_{loc}^l, \boldsymbol{z}_{glo}^l\right] = \mathrm{GELU}\left(\boldsymbol{W}_p \mathrm{LN}\left(\tilde{\boldsymbol{f}}^l\right)\right) \tag{3}$$

where $\boldsymbol{W}_p$ denotes the linear projection matrix and GELU$(\cdot)$ and LN$(\cdot)$ refer to the GELU nonlinearity and layer normalization, respectively. In addition, spatial global averaging over $\boldsymbol{z}_{glo}^l$ yields the scene-level semantic representation $\boldsymbol{g}^l$

$$\boldsymbol{g}^l = \frac{1}{h^l w^l} \sum_{n=1}^{h^l w^l} \boldsymbol{z}_{glo,n}^l \tag{4}$$

where h^l and w^l correspond to the height and width of the feature map, respectively.

Furthermore, the global semantic descriptor $\boldsymbol{g}^l$ is expanded and concatenated with each local feature $\boldsymbol{z}_{loc}^l$ so that the local representations can incorporate richer contextual information. The final saliency score is then produced by a feed-forward network

$$s^l = \sigma\left(\mathcal{F}_{out}\left(\mathrm{LN}\left(\boldsymbol{z}_{loc}^l \parallel \boldsymbol{g}^l\right)\right)\right) \tag{5}$$

where $\sigma(\cdot)$ refers to the Sigmoid function, $\mathcal{F}_{out}$ is the feed-forward network, and $\parallel$ indicates channel concatenation.

2) Sparse Feature Interaction and Feature Fusion: Upon obtaining the predicted saliency scores for each feature level, the SGSE executes dynamic sparse sampling based on these scores. Specifically, it first filters the indices of key features corresponding to the top $\rho\%$ of saliency scores layer by layer to construct the highly responsive feature index set Ω^l

$$\Omega^l = \left\{n \mid s_n^l \geq \tau_\rho^l\right\}, \quad l < L \tag{6}$$

where τ_ρ^l denotes the lower bound of the top $\rho\%$ saliency scores at the lth level.

As illustrated in Fig. 3, shallow and deep features are processed with different attention mechanisms. Specifically, shallow features ($l < L$) retain richer local spatial details, and the features within Ω^l are selected as queries for deformable attention to update target-relevant representations, whereas the unselected features are directly preserved through an identity mapping. In contrast, for deep features ($l = L$), a self-attention mechanism is adopted to enhance semantic representation, owing to their lower spatial resolution and more compact semantics. Following these attention updates, the resulting feature $\hat{\boldsymbol{f}}_n^l$ is formulated as follows:

$$\hat{\boldsymbol{f}}_n^l = \begin{cases} \mathrm{MHSA}\left(\boldsymbol{f}_n^l, \boldsymbol{F}^l\right), & l = L \\ \mathrm{MHDA}\left(\boldsymbol{f}_n^l, \boldsymbol{p}_n, \boldsymbol{F}^l\right), & l \neq L \wedge n \in \Omega^l \\ \boldsymbol{f}_n^l, & l \neq L \wedge n \notin \Omega^l \end{cases} \tag{7}$$

where $\boldsymbol{p}_n = (x_n, y_n) \in \mathbb{R}^2$ represents the spatial coordinates corresponding to $\boldsymbol{f}_n^l$, MHSA$(\cdot,\cdot)$ denotes multihead self-attention, and MHDA$(\cdot,\cdot,\cdot)$ indicates multihead deformable attention [48].

Deformable attention is, therefore, used as the update operator within a saliency-conditioned asymmetric encoding strategy: selected shallow tokens are updated, unselected shallow tokens are preserved, and deep tokens retain full semantic interaction.

Upon completion of the sparse feature interaction, the updated feature vector $\hat{\boldsymbol{f}}^l$ is reconstructed into the feature map $\hat{\boldsymbol{F}}^l$ via dimension transformation. Subsequently, the CCFF module [31] is employed to aggregate multiscale features,

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enhancing their contextual representation capabilities and, thereby, yielding the fused features $\hat{\boldsymbol{O}}^l$

$$\left\{\hat{\boldsymbol{O}}^l\right\}_{l=1}^{L} = \mathrm{CCFF}\left(\left\{\hat{\boldsymbol{F}}^l\right\}_{l=1}^{L}\right). \tag{8}$$

Building upon this, to ensure that the decoder focuses on potential target regions and effectively suppresses background clutter interference, the predicted saliency map S^l is further utilized as a soft attention mask to modulate the fused features $\hat{\boldsymbol{O}}^l$, ultimately producing the output features $\boldsymbol{O}^l$

$$\boldsymbol{O}^l = \hat{\boldsymbol{O}}^l \odot S^l \tag{9}$$

where $S^l = \mathcal{R}(s^l)$. Sparse interaction limits the propagation of clutter-dominated shallow responses during attention-based updating, while soft attention mask modulation attenuates residual background responses introduced during multiscale feature aggregation. Together, they produce the output features $\{\boldsymbol{O}^l\}_{l=1}^{L}$ while retaining the multiscale context needed by the decoder.

3) Saliency Loss: The HSN is trained utilizing a saliency loss. The saliency labels are generated from bounding-box supervision to provide target-related foreground guidance at each feature level. Specifically, for the lth feature map $\boldsymbol{F}^l$ with downsampling factor k_l, a spatial position (i, j) on the feature map corresponds to the coordinate (x, y) in the original image as follows:

$$x = \left\lfloor \frac{k^l}{2} \right\rfloor + i \cdot k^l, \quad y = \left\lfloor \frac{k^l}{2} \right\rfloor + j \cdot k^l. \tag{10}$$

By determining whether (x, y) falls within the extent of the foreground targets, the hierarchical saliency training label $M_{i,j}^l$ can be obtained

$$M_{i,j}^l = \begin{cases} 1, & (x, y) \in \mathcal{D}_{Bbox} \\ 0, & (x, y) \notin \mathcal{D}_{Bbox} \end{cases} \tag{11}$$

where $\mathcal{D}_{Bbox}$ represents the set of all foreground regions within the SAR image.

Based on the above, the saliency loss for the HSN is formulated as a focal loss [56] over the predicted saliency scores and the corresponding saliency training labels

$$\mathcal{L}_s = \sum_{l=1}^{L} \sum_{n=1}^{h^l w^l} -\alpha \left(1 - p_n^l\right)^{\gamma} \log\left(p_n^l\right) \tag{12}$$

where α and γ are both hyperparameters of the focal loss and p_n^l represents the estimated probability for the ground truth.

C. Group Policy Optimization Decoder

GPOD parameterizes each predicted box as a Gaussian distribution. During training, its supervised branch constrains the predicted means, while its reinforcement learning branch samples candidate-box sets from the joint policy and optimizes their image-level localization reward. At inference, only the predicted means are retained.

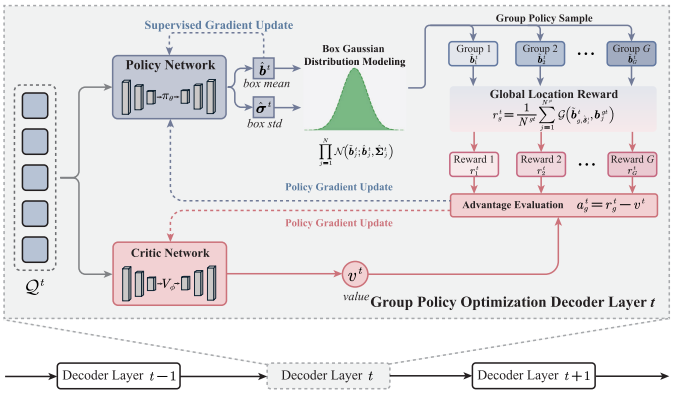


Fig. 4. GPOD. During training, GPOD models each predicted bounding box as a Gaussian distribution and samples candidate boxes for localization optimization. The supervised and reinforcement learning branches are jointly optimized to improve localization robustness under ambiguous target extents.

1) Group Policy Sampling: As illustrated in Fig. 4, the decoder is formed by the cascaded stacking of multiple decoder layers. Using the tth decoder layer for illustration, its input consists of query features $\mathcal{Q}^t = \{\boldsymbol{q}_j^t\}_{j=1}^{N}$, with N indicating the number of queries. The layer primarily comprises a policy network and a critic network. Specifically, the GPOD module integrates two branches: supervised learning and reinforcement learning. The supervised learning branch corresponds to the policy network and is trained under the constraints of the standard RT-DETR supervised loss. The reinforcement learning branch is composed of both the policy and critic networks, supervised by the corresponding reinforcement learning loss. Details of these components are given in the following.

To map the latent query features into a spatial probability distribution of target bounding boxes, a policy network π_θ is designed. Its output layer adopts two parallel MLP branches, which estimate the mean and standard deviation parameters of the Gaussian distribution associated with the target box, respectively, given by

$$\left[\hat{\boldsymbol{b}}_j^t, \hat{\boldsymbol{\sigma}}_j^t\right] = \pi_\theta\left(\boldsymbol{q}_j^t\right) \quad \forall j \in \{1, \ldots, N\} \tag{13}$$

where $\hat{\boldsymbol{b}}_j^t = [\hat{x}_j^t, \hat{y}_j^t, \hat{w}_j^t, \hat{h}_j^t]^\top$ represents the estimated target boxes and $\hat{\boldsymbol{\sigma}}_j^t = [\hat{\sigma}_{x,j}^t, \hat{\sigma}_{y,j}^t, \hat{\sigma}_{w,j}^t, \hat{\sigma}_{h,j}^t]^\top$ represents the standard deviations corresponding to each coordinate.

Assuming that the target instances corresponding to each query feature $\boldsymbol{q}_j^t$ in the tth layer satisfy the independent and identically distributed condition, the joint distribution of the N predicted bounding boxes is defined as

$$\pi_\theta\left(\tilde{\mathcal{B}}^t \mid \mathcal{Q}^t\right) = \prod_{j=1}^{N} \mathcal{N}\left(\tilde{\boldsymbol{b}}_j^t; \hat{\boldsymbol{b}}_j^t, \hat{\boldsymbol{\Sigma}}_j^t\right) \tag{14}$$

where $\tilde{\mathcal{B}}^t$ denotes the collection of predicted target boxes at the tth layer and $\hat{\boldsymbol{\Sigma}}_j^t = \mathrm{diag}((\hat{\boldsymbol{\sigma}}_j^t)^2)$ denotes the covariance matrix of the jth prediction result, which is a diagonal matrix.

After constructing the joint Gaussian distribution for potential target bounding boxes, a single sampled prediction set provides only one stochastic realization of the candidate space and can yield a high-variance policy-gradient estimate. GPOD, therefore, draws multiple groups of complete prediction sets

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from the same policy. These samples explore alternative box extents around the current predictions and provide a more stable estimate of the expected image-level localization reward. The group sampling strategy is formulated as

$$\left\{\tilde{\mathcal{B}}_g^t\right\}_{g=1}^{G} \overset{\text{i.i.d.}}{\sim} \pi_\theta\left(\tilde{\mathcal{B}}^t \mid \mathcal{Q}^t\right) \tag{15}$$

where G is the number of sampling groups, $\tilde{\mathcal{B}}_g^t = \{\tilde{\boldsymbol{b}}_{g,j}^t\}_{j=1}^N$ represents the bounding boxes obtained in the gth sampling group, and $\tilde{\boldsymbol{b}}_{g,j}^t$ is the jth bounding box sampled from the tth decoder layer in the gth group.

2) Supervised-Reinforcement Collaborative Optimization: Although prediction quality can be assessed by computing the overlap between sampled boxes and the corresponding ground-truth boxes, DETR-based box prediction may assign the same box distribution to different ground-truth targets across different sampling groups, which, in turn, introduces inconsistency into training. To address this, GPOD employs a CBM strategy to decouple the allocation optimization process from the stochastic evaluation process, ensuring that a specific bounding box distribution consistently corresponds to the same ground-truth target regardless of the sampling outcomes. Specifically, the Hungarian algorithm is applied to the ground-truth set, and the prediction results are composed only of classification scores $\hat{\boldsymbol{c}}^t$ and bounding box means $\hat{\boldsymbol{b}}^t$ to derive a global optimal permutation $\hat{\boldsymbol{\delta}}^t$ shared across all G sampling groups

$$\hat{\boldsymbol{\delta}}^t = \operatorname{argmin}_{\boldsymbol{\delta}^t \in \mathfrak{S}_N} \sum_{j=1}^{N} \mathcal{L}_{\text{match}}\left(\boldsymbol{y}_j, \hat{\boldsymbol{y}}_{\boldsymbol{\delta}_j^t}^t\right) \tag{16}$$

where $\hat{\boldsymbol{y}}_{\boldsymbol{\delta}_j^t}^t = (\hat{\boldsymbol{c}}_{\boldsymbol{\delta}_j^t}^t, \hat{\boldsymbol{b}}_{\boldsymbol{\delta}_j^t}^t)$ represents the prediction result with index $\hat{\boldsymbol{\delta}}_j^t$.

Since the supervised branch already provides instancewise classification and box-regression gradients, GPOD introduces an image-level localization reward to evaluate each complete sampled prediction set. Specifically, given $\hat{\boldsymbol{\delta}}^t$, the reward r_g^t for the gth sampling group is defined as the average normalized GIoU [57] over all matched targets in the image

$$r_g^t = \frac{1}{N^{gt}} \sum_{j=1}^{N^{gt}} \mathcal{G}\left(\tilde{\boldsymbol{b}}_{g,\hat{\boldsymbol{\delta}}_j^t}^t, \boldsymbol{b}_j^{gt}\right) \tag{17}$$

where N^{gt} indicates the total count of ground-truth instances and $\mathcal{G}(\tilde{\boldsymbol{b}}_{g,\hat{\boldsymbol{\delta}}_j^t}^t, \boldsymbol{b}_j^{gt})$ gives the normalized GIoU score between the sampled prediction and the matched ground-truth box

$$\mathcal{G}\left(\tilde{\boldsymbol{b}}_{g,\hat{\boldsymbol{\delta}}_j^t}^t, \boldsymbol{b}_j^{gt}\right) = \frac{1 + \text{GIoU}\left(\tilde{\boldsymbol{b}}_{g,\hat{\boldsymbol{\delta}}_j^t}^t, \boldsymbol{b}_j^{gt}\right)}{2}. \tag{18}$$

The supervised branch already optimizes a GIoU term within the standard RT-DETR box-regression loss. GPOD, therefore, adopts normalized GIoU as the reinforcement learning reward so that the supervised regression objective and the policy objective evaluate localization using the same overlap criterion. This alignment encourages compatible localization updates from the two optimization branches.

For comparison, the instance-level reward assigned to the jth matched target in the gth sampling group is defined as

$$r_{g,j}^{t,\text{ins}} = \mathcal{G}\left(\tilde{\boldsymbol{b}}_{g,\hat{\boldsymbol{\delta}}_j^t}^t, \boldsymbol{b}_j^{gt}\right), \quad j = 1, \ldots, N^{gt}. \tag{19}$$

Each $r_{g,j}^{t,\text{ins}}$ independently weights the log-probability of its corresponding sampled box. In contrast, the image-level reward r_g^t averages these instance-level scores over all matched targets and applies the resulting advantage to the joint log-probability of the complete sampled prediction set. This image-level credit assignment complements the instancewise supervised loss by jointly optimizing the localization quality of all matched predictions.

For the reinforcement learning branch of GPOD, let the critic network V_ϕ output a value $v^t = V_\phi(\mathcal{Q}^t)$. Its training objective is defined as follows:

$$\mathcal{J}^t(\theta) = \mathbb{E}_{\{\tilde{\mathcal{B}}_g^t\}_{g=1}^{G} \sim \pi_\theta(\cdot|\mathcal{Q}^t)} \left[\frac{1}{G} \sum_{g=1}^{G} \log \pi_\theta\left(\tilde{\mathcal{B}}_g^t \mid \mathcal{Q}^t\right) \cdot a_g^t\right] \tag{20}$$

where $a_g^t = r_g^t - v^t$ represents the advantage function corresponding to the gth sampling result, computed by subtracting the value estimate v^t from the reward r_g^t.

The loss function for the critic network is defined as

$$\mathcal{L}_{\text{value}}^t(\phi) = \frac{1}{G} \sum_{g=1}^{G} \left\| r_g^t - v^t \right\|^2. \tag{21}$$

Ultimately, the tth decoder layer is optimized with a total loss $\mathcal{L}_{\text{decoder}}^t$, which combines the supervised training objective with the reinforcement learning objective

$$\mathcal{L}_{\text{decoder}}^t = \mathcal{L}_{\text{sup}}^t + \lambda_{\text{rl}}\left(-\lambda_{\text{policy}} \mathcal{J}^t(\theta) + \mathcal{L}_{\text{value}}^t(\phi)\right) \tag{22}$$

where λ_{policy} and λ_{rl} are weighting coefficients and $\mathcal{L}_{\text{sup}}^t$ denotes the standard RT-DETR loss.

D. Model Optimization and Inference

The training objective of SP-DETR consists of two components, namely, the SGSE-derived saliency loss $\mathcal{L}_s$ and the GPOD decoder loss $\mathcal{L}_{\text{decoder}}^t$

$$\mathcal{L}_{\text{total}} = \lambda_s \mathcal{L}_s + \sum_t \mathcal{L}_{\text{decoder}}^t \tag{23}$$

where λ_s is the weight for the saliency loss $\mathcal{L}_s$. Since the RT-DETR model typically comprises six decoder layers but only one encoder layer, λ_s is set to 6 to balance the losses between the encoder and the decoder.

In the inference phase, the standard deviation branch within the policy network and the critic network in the GPOD can be removed; therefore, this component introduces no additional computational overhead during inference. Furthermore, the mean $\hat{\boldsymbol{b}}^t$ predicted by the policy network is directly utilized as the predicted bounding box.

IV. Experiments

A. Dataset

To assess the performance of SP-DETR, we conduct experiments on two public SAR object detection benchmarks, namely, SAR-AIRCraft-1.0 [4] and SARDet-100K [6]. In addition, ablation experiments are conducted to examine the contribution of each component in the proposed framework. The dataset scale, including data volume and target annotations, is reported in Table I, and the category distribution together with representative samples is shown in Fig. 5.

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TABLE I
DATASET SCALE STATISTICS OF SAR-AIRCRAFT-1.0 AND SARDET-100K

Dataset	Images				Instances				Ins/Img
	Train	Val	Test	All	Train	Val	Test	All	
SAR-AIRCraft-1.0 [4]	3,047	442	879	4,368	11,551	1,962	2950	16,463	3.74
SARDet-100K [6]	94,493	10,492	11,613	116,598	198,747	22,703	24,023	245,653	2.11

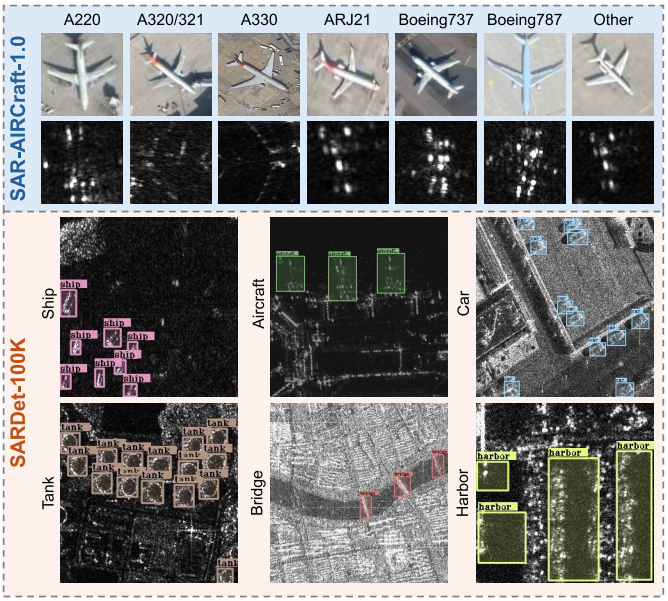


Fig. 5. Target categories and representative image samples of SAR-AIRCraft-1.0 and SARDet-100K datasets.

1) SAR-AIRCraft-1.0: SAR-AIRCraft-1.0 [4] is constructed for fine-grained aircraft detection in SAR imagery. The data were acquired by the Gaofen-3 satellite under spotlight imaging mode using HH polarization with a 1-m resolution. Its imaging scenes span several civil airports, such as Shanghai Hongqiao Airport, Beijing Capital International Airport, and Taiwan Taoyuan International Airport. In total, the dataset provides 4368 SAR images and 16 463 aircraft annotations across seven fine-grained categories, including A220, A320/321, A330, ARJ21, Boeing737, Boeing787, and Other.[1]

2) SARDet-100K: SARDet-100K [6] serves as one of the largest benchmarks for SAR object detection. It aggregates a wide range of publicly available SAR detection datasets and standardizes their heterogeneous formats through data cleaning. In total, it contains about 117k SAR images and 246k target annotations from six object categories, namely, Ship, Airplane, Car, Tank, Bridge, and Harbor. Since the data are collected from diverse sources, SARDet-100K involves substantial resolution variations as well as different frequency bands and polarization modes, which increases the difficulty of robust SAR target detection.

B. Implementation Details

Both the comparison experiments and the ablation studies are implemented on the MMDetection platform under a unified evaluation protocol. For SAR-AIRCraft-1.0 [4], following the preprocessing strategy of DenoDet [40], each image is divided into 512 × 512 patches by a sliding-window scheme with a stride of 200 for both training and testing. During evaluation, the predictions obtained from all cropped patches are merged back to the corresponding original image. The model is trained for 12 epochs using the AdamW optimizer, with the learning rate and weight decay both set to 10^{-4} and the batch size fixed at 16. After the first six epochs, the learning rate is further adjusted using a cosine annealing schedule.

On SARDet-100K [6], the input resolution is fixed to 512 × 512. Training is performed for 72 epochs with AdamW, where the initial learning rate is 10^{-4}. After epoch 36, cosine annealing is applied to decay the learning rate progressively.

For the comparative methods, to ensure the fairness of the comparative experiments, ResNet-50 [15] is uniformly used as the backbone network of the corresponding models.[2]

C. Evaluation Metric

Model performance is assessed using mean Average Precision (mAP) together with its commonly used variants. For a predefined IoU threshold set Φ_{IoU}, the mAP is defined as

$$\text{mAP} = \frac{1}{|C| \cdot |\Phi_{\text{IoU}}|} \sum_{c \in C} \sum_{\phi \in \Phi_{\text{IoU}}} \text{AP}_c(\phi) \tag{24}$$

where C is the set of categories, $\sharp$ gives the number of elements in a set, and $\text{AP}_c(\phi)$ is the average precision for category c at a given IoU value ϕ. By setting different confidence thresholds τ, varying precision $P_{c,\phi}(\tau)$ and recall $R_{c,\phi}(\tau)$ can be obtained to form the P–R curve, with $\text{AP}_c(\phi)$ representing the area under the P–R curve

$$\text{AP}_c(\phi) = \int_0^1 P_{c,\phi}\left(R_{c,\phi}\right) \mathrm{d}R_{c,\phi}. \tag{25}$$

By further setting the IoU threshold sets Φ_{IoU} to $\Phi_{\text{IoU}} = \{0.5\}$, $\Phi_{\text{IoU}} = \{0.75\}$, and $\Phi_{\text{IoU}} = \{0.5, 0.55, \ldots, 0.95\}$, mAP_{50}, mAP_{75}, and $\text{mAP}_{50:95}$ can be calculated, respectively. The three variants characterize complementary aspects of detection performance. mAP_{50} reflects detection under a relatively permissive localization criterion, mAP_{75} is more sensitive to precise localization, and $\text{mAP}_{50:95}$ provides a comprehensive evaluation across progressively stricter IoU criteria.

D. Comparison With Existing Methods

We evaluate SP-DETR on SAR-AIRCraft-1.0 [4] and SARDet-100K [6] through extensive comparisons with existing detectors. For comparison, we consider representative

[1] RF-DETR uses a DINOv2-pretrained ViT-Small backbone.

[2] YOLO26-Large is used.

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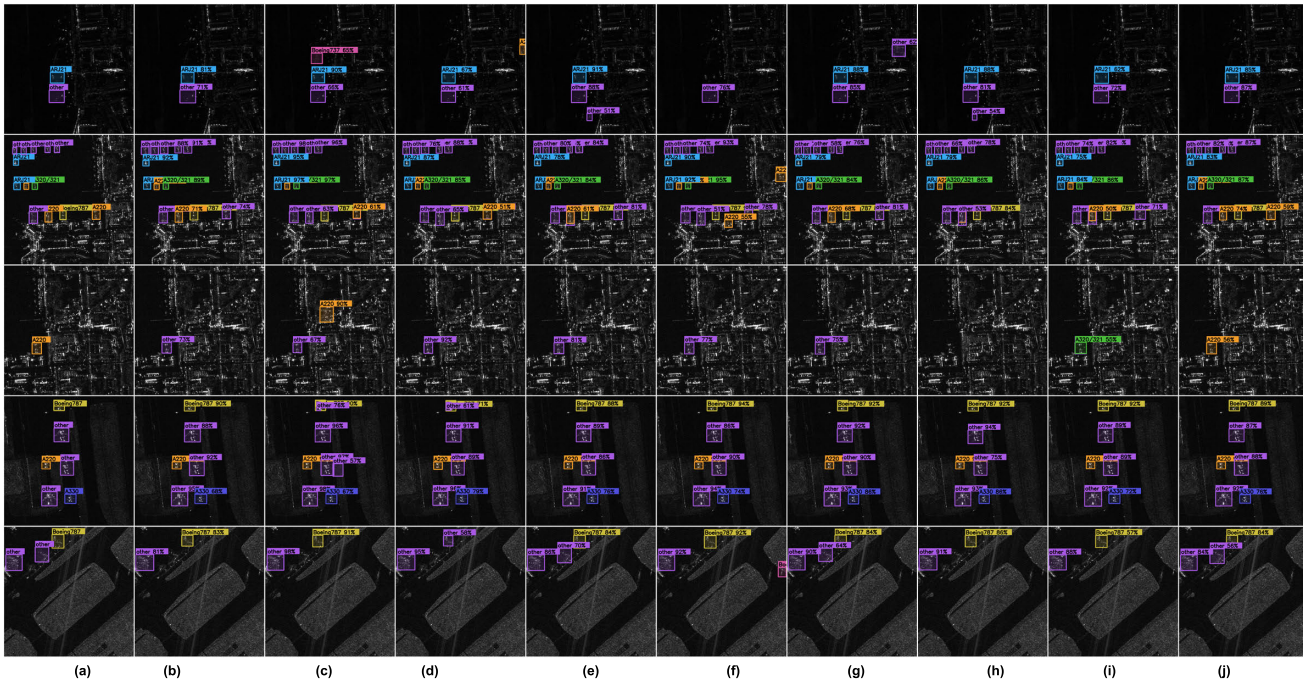


Fig. 6. Detection visualization on SAR-AIRCraft-1.0 [4] for representative end-to-end methods. Correct predictions are shown with green boxes, false alarms with red boxes, and missed objects with yellow circles. All results are obtained using a confidence threshold of 0.5. (a) GT. (b) Deformable DETR. (c) DAB-DETR. (d) Conditional DETR. (e) RT-DETR. (f) DINO. (g) DFINE. (h) DEIM. (i) RF-DETR. (j) SP-DETR(Ours).

detectors with publicly available implementations or reported results on the adopted datasets. The comparison protocols include 26 representative methods on SAR-AIRCraft-1.0 and 28 on SARDet-100K, spanning one-stage, two-stage, and end-to-end detectors. The one-stage methods common to both datasets are [37], [40], [56], [58], [59], [60], [61], [62], [63], [64], [65], and [66], and the two-stage methods are [17], [67], [68], [69], and [70]. The end-to-end methods include DETR [28], Deformable DETR [48], DAB-DETR [49], Conditional DETR [71], RT-DETR [31], DINO [50], DFINE [55], DEIM [34], and RF-DETR [72]. For SARDet-100K, we additionally include MD-DETR [41], which is specifically designed for SAR images, and YOLO26 [73].

Table II reports the comparison results on SAR-AIRCraft-1.0 [4]. SP-DETR achieves the highest scores across the main evaluation metrics, with $AP_{50:95}$, AP_{50}, and AP_{75} reaching 51.35, 69.47, and 56.44, respectively. Relative to the RT-DETR baseline, SP-DETR brings a clear accuracy gain while introducing only marginal additional parameters and computational overhead, indicating a favorable tradeoff between detection accuracy and inference efficiency. Notably, its AP_{75} is even higher than that of more complex two-stage detectors, such as Cascade R-CNN [67] and Dynamic R-CNN [68], suggesting stronger localization capability. The qualitative results in Fig. 6 further compare different end-to-end detectors on SAR-AIRCraft-1.0, where SP-DETR produces more accurate detection results than the competing end-to-end methods.

The results on the larger SARDet-100K [6] benchmark are summarized in Table III. SP-DETR achieves the best overall performance in terms of $AP_{50:95}$ and AP_{50}, reaching 61.53 and 90.88, respectively, and improves upon the RT-DETR baseline by 1.15 $AP_{50:95}$ and 0.55 AP_{50}. Although RF-DETR attains a higher AP_{75} score (65.74 versus 64.65), SP-DETR maintains a higher overall $AP_{50:95}$, indicating stronger performance when evaluated across multiple IoU thresholds. The visual comparisons in Fig. 7 provide representative detection results of the end-to-end methods on SARDet-100K.

To provide a more intuitive comparison, Fig. 8 presents bubble charts constructed from the performance of different models on SAR-AIRCraft-1.0 and SARDet-100K. In each chart, $AP_{50:95}$ is used as the x-axis, AP_{50} is used as the y-axis, and the bubble area indicates the FLOPs of each detector. As shown in Fig. 8, SP-DETR provides a favorable accuracy–complexity tradeoff among the compared methods.

E. Ablation Analysis

We further evaluate SP-DETR on both datasets through ablation experiments to analyze the effectiveness of its individual modules.

First, the SGSE module is individually added to the baseline RT-DETR. This module enhances target-related feature representation through saliency prediction and sparse feature encoding. The third row of Tables IV and V shows that SGSE effectively improves detection performance on both datasets. Specifically, on SAR-AIRCraft-1.0 [4], the $AP_{50:95}$, AP_{50}, and AP_{75} scores increase from 45.22 to 48.48, 62.67 to 65.72, and 49.67 to 52.53, respectively. On SARDet-100K [6], SGSE also raises $AP_{50:95}$, AP_{50}, and AP_{75} from 60.38 to 61.02, 90.33 to 90.69, and 63.86 to 64.49, respectively.

We then evaluate the individual contribution of GPOD by adding it alone to the RT-DETR baseline. The fourth rows of Tables IV and V indicate that GPOD improves the detection performance on both datasets. For SAR-AIRCraft-1.0, $AP_{50:95}$, AP_{50}, and AP_{75} increase from 45.22, 62.67, and 49.67 to 49.03, 65.65, and 52.71, respectively. For SARDet-100K, the

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TABLE II
COMPARATIVE PERFORMANCE ON SAR-AIRCRAFT-1.0 [4] WITH AP_{50} REPORTED FOR EACH CATEGORY

Method	FLOPs	Params	$AP_{50:95}$	AP_{50}	AP_{75}	A220	A320	A330	ARJ21	B737	B787	Other
One-Stage												
FCOS [58]	51.58G	32.13M	45.07	62.40	50.52	59.21	86.14	92.47	57.76	35.43	47.33	58.48
GFL [37]	52.37G	32.27M	47.47	64.55	53.21	62.03	88.64	97.68	58.10	34.47	48.32	62.62
RepPoints [59]	48.50G	36.82M	46.26	65.21	52.72	56.09	91.22	90.91	57.25	38.23	54.23	68.54
ATSS [60]	51.58G	32.13M	46.89	62.67	51.03	55.19	88.73	95.26	59.41	34.21	49.39	56.47
CenterNet [61]	51.57G	32.13M	47.64	63.26	51.26	65.15	86.11	96.81	53.87	35.71	46.58	58.57
PAA [62]	51.58G	32.13M	49.06	67.06	52.42	60.22	91.60	96.20	64.06	38.75	56.95	61.61
RetinaNet [56]	52.88G	36.45M	47.38	66.28	53.08	67.90	89.47	93.79	62.36	35.76	53.46	61.20
TOOD [63]	50.53G	32.03M	47.16	63.03	51.15	57.22	91.30	94.64	53.82	33.57	51.46	59.19
VFNet [64]	48.39G	32.72M	49.21	66.31	54.72	61.32	93.15	94.31	60.70	33.12	61.67	59.88
YOLOF [65]	26.33G	42.48M	45.28	66.51	50.29	62.76	89.60	92.01	60.05	39.00	56.66	65.46
AutoAssign [66]	51.84G	36.26M	46.90	64.57	49.05	57.56	89.77	96.96	61.80	35.99	45.62	64.31
DenoDet [40]	48.53G	70.33M	47.16	68.60	54.70	64.82	90.54	93.88	59.58	46.32	56.04	69.05
Two-Stage												
Faster RCNN [17]	63.21G	41.37M	45.61	62.93	52.96	48.67	89.73	89.50	53.95	33.27	59.57	65.81
Cascade RCNN [67]	91.00G	69.17M	48.53	65.33	56.11	58.71	97.93	97.12	53.04	32.21	57.12	61.16
Dynamic RCNN [68]	63.21G	41.37M	48.06	65.59	56.01	58.69	93.61	96.88	53.65	37.42	54.45	64.41
Grid RCNN [69]	0.18T	64.47M	45.62	62.63	51.48	52.95	89.29	94.80	61.58	32.32	44.51	62.93
Libra RCNN [70]	64.02G	41.64M	47.27	64.91	53.95	49.60	94.34	95.29	60.77	32.37	57.22	64.80
End2end												
DETR [28]	24.94G	41.56M	3.40	8.49	1.75	14.79	0.33	0.01	5.81	5.38	17.94	15.20
Deformable DETR [48]	51.78G	40.10M	45.62	67.03	49.98	64.47	90.70	96.25	57.03	38.19	59.23	63.34
DAB-DETR [49]	28.94G	43.70M	35.55	53.68	36.75	55.37	85.40	26.90	54.20	39.36	53.11	61.41
Conditional DETR [71]	28.09G	43.45M	41.49	64.29	44.62	61.17	84.40	86.07	63.50	38.38	58.35	58.16
RT-DETR [51]	43.99G	42.78M	45.22	62.67	49.67	58.84	80.09	85.32	62.37	33.96	57.93	60.18
DINO [50]	80.72G	47.55M	46.93	65.92	48.86	75.01	88.83	79.70	62.27	32.66	56.94	66.02
DFINE [55]	35.12G	37.19M	46.72	64.16	51.45	59.03	89.25	94.89	64.20	32.80	51.98	56.96
DEIM [34]	35.12G	37.19M	45.93	61.85	49.33	52.44	88.88	90.70	55.01	34.20	51.36	60.38
RF-DETR[1] [72]	78.14G	31.87M	42.63	64.19	50.24	65.87	86.64	96.92	57.36	27.31	47.36	67.86
SP-DETR (Ours)	47.02G	44.95M	**51.35**	**69.47**	**56.44**	70.09	94.81	93.14	67.39	43.45	53.09	64.34

corresponding metrics are improved from 60.38, 90.33, and 63.86 to 60.76, 90.55, and 64.04.

Finally, SGSE and GPOD are jointly integrated into RT-DETR. As reported in the fifth rows of Tables IV and V, the complete SP-DETR model reaches 51.35 $AP_{50:95}$, 69.47 AP_{50}, and 56.44 AP_{75} on SAR-AIRCraft-1.0 and 61.53 $AP_{50:95}$, 90.88 AP_{50}, and 64.65 AP_{75} on SARDet-100K. This indicates that SGSE and GPOD provide complementary improvements to the baseline.

F. Analysis

This section provides further analysis of the proposed SGSE and GPOD modules.

1) Analysis of SGSE: Shallow high-resolution features preserve richer local spatial details, but dense attention over all shallow tokens incurs substantial computational cost and can be affected by background interference. SGSE addresses this issue by using saliency scores to select target-relevant tokens for sparse interaction. To analyze the effect of different attention operators, we keep the saliency-guided sparse encoding strategy unchanged and replace only the attention operator with multihead self-attention or deformable attention. The results in Table VI show that deformable attention on selected shallow features improves detection accuracy while maintaining better computational efficiency.

Table VII further decomposes SGSE through three independent variants under the complete SP-DETR setting. In the table, CLG, SI, and SAMM denote cross-level saliency guidance, sparse interaction, and soft attention mask modulation, respectively. The auxiliary saliency supervision (Aux. Sup.) variant retains HSN and its auxiliary saliency loss but prevents the predicted saliency from affecting the detection forward path. Its lower performance shows that auxiliary supervision alone cannot reproduce the benefit of SGSE. Removing SAMM while retaining hierarchical prediction and SI reduces $AP_{50:95}$, AP_{50}, and AP_{75} by 1.66, 1.53, and 2.56 points, respectively, confirming the contribution of soft attention mask modulation. Setting $\beta^l = 0$ to remove CLG causes reductions of 0.34, 0.60, and 2.14 points. The larger decrease in AP_{75} indicates that cross-level saliency guidance mainly benefits stricter localization.

Furthermore, SGSE performs sparse encoding by selecting only the top $\rho\%$ of each shallow feature layer based on the predicted saliency scores. To analyze how this selection ratio affects detection accuracy, we use RT-DETR with GPOD as the baseline and vary ρ from 10 to 60 with an interval of 10. As shown in Fig. 9, increasing ρ initially improves performance, but further enlargement leads to a decline. A possible reason is that retaining too many tokens introduces additional background interference into the foreground representation, which

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TABLE III

COMPARATIVE PERFORMANCE ON THE SARDET-100K [6] WITH $AP_{50:95}$ REPORTED FOR EACH CATEGORY

Method	FLOPs	Params	$AP_{50:95}$	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Ship	Aircraft	Car	Tank	Bridge	Harbor
One-stage														
FCOS [58]	51.57G	32.13M	52.52	85.82	54.93	47.01	66.13	57.82	59.79	55.44	60.75	41.78	34.17	63.44
GFL [37]	52.36G	32.27M	55.01	85.16	58.87	49.44	67.29	60.45	63.92	57.63	62.29	44.80	36.41	65.04
RepPoints [59]	48.49G	36.82M	51.66	86.43	53.99	46.66	63.26	53.78	60.85	55.50	61.13	40.69	35.12	56.71
ATSS [60]	51.57G	32.13M	54.95	87.60	58.25	49.89	67.94	58.97	61.53	55.94	61.77	46.20	37.22	67.48
CenterNet [61]	51.57G	32.13M	53.91	86.17	57.31	48.88	66.22	57.74	61.24	56.35	61.74	45.31	35.91	63.29
PAA [62]	51.57G	32.13M	52.20	85.71	54.80	46.00	63.90	57.61	60.16	56.17	60.09	41.07	35.96	60.12
RetinaNet [56]	52.77G	36.43M	46.48	77.74	48.94	40.25	59.35	50.26	55.36	54.00	60.88	32.72	24.81	51.12
TOOD [63]	50.52G	30.03M	54.65	86.88	58.41	50.20	66.72	58.60	62.28	55.61	62.53	45.96	36.64	65.24
VFNet [64]	48.38G	32.72M	53.01	84.32	56.32	47.37	65.39	57.99	62.14	55.84	61.97	42.08	34.11	62.28
YOLOF [65]	26.32G	42.46M	42.83	74.95	43.18	33.73	56.19	53.57	52.62	52.64	52.71	22.86	23.74	52.42
AutoAssign [66]	51.83G	36.26M	53.95	89.58	55.96	50.14	63.40	54.73	62.03	55.70	61.69	48.55	38.25	57.45
YOLO26[2] [73]	59.62G	26.19M	50.88	81.38	55.30	47.09	64.04	54.70	62.32	43.35	66.18	38.08	36.00	60.44
DenoDet [40]	52.69G	70.33M	55.88	85.81	60.16	50.63	68.47	60.96	64.91	57.36	63.66	45.79	36.39	67.17
Two-stage														
Faster RCNN [17]	63.21G	41.37M	39.22	70.04	39.87	32.55	47.23	42.02	50.45	50.36	57.82	24.90	18.69	33.11
Cascade RCNN [67]	90.99G	69.17M	53.55	87.33	56.81	49.09	62.89	48.68	66.99	56.43	63.25	44.35	36.89	53.81
Dynamic RCNN [68]	63.21G	41.37M	49.75	80.96	53.91	43.12	59.72	54.77	61.32	53.86	60.00	33.68	34.40	55.25
Grid RCNN [69]	0.18T	64.47M	50.05	80.58	53.49	42.43	62.01	52.70	60.43	55.61	61.94	36.03	31.16	55.13
Libra RCNN [70]	64.02G	41.64M	52.09	83.54	55.81	45.85	63.52	55.40	61.32	54.03	61.56	38.12	35.97	61.50
End2end														
DETR [28]	24.94G	41.56M	45.73	78.57	46.87	37.01	58.16	55.58	54.94	51.17	50.11	26.06	32.80	59.31
Deformable DETR [48]	51.78G	40.10M	52.00	88.77	54.03	46.99	63.58	58.55	60.94	54.16	61.22	39.14	36.09	60.46
DAB-DETR [49]	28.94G	43.70M	43.31	78.14	43.10	34.82	56.34	52.62	53.16	50.32	49.47	24.06	28.47	55.07
Conditional DETR [71]	28.09G	43.45M	44.04	77.88	44.40	35.25	56.47	52.86	52.77	49.58	51.00	22.73	29.98	58.16
RT-DETR [51]	43.99G	42.78M	60.38	90.33	63.86	56.55	72.61	66.01	72.07	48.10	66.76	52.23	47.42	76.18
DINO [50]	80.72G	47.55M	56.94	89.96	59.31	52.27	71.03	65.14	68.20	45.96	63.65	45.03	42.20	77.15
DFINE [55]	35.12G	37.19M	60.43	89.26	63.80	57.24	73.50	65.78	72.61	47.91	65.61	51.93	47.14	77.96
DEIM [34]	35.12G	37.19M	58.44	88.09	60.95	55.17	71.43	63.17	72.00	46.48	65.39	50.67	44.32	72.34
RF-DETR [72]	78.14G	31.87M	59.75	90.70	**65.74**	55.86	69.90	67.46	68.69	46.30	67.79	50.95	47.17	77.97
MD-DETR [41]	-	-	57.20	88.50	62.80	50.30	68.30	63.80	-	-	-	-	-	-
SP-DETR (Ours)	47.02G	44.95M	**61.53**	**90.88**	64.65	58.95	73.21	67.18	72.33	45.48	66.47	54.51	49.78	80.75

TABLE IV

ABLATION ANALYSIS ON SAR-AIRCRAFT-1.0 [4], WITH AP_{50} REPORTED FOR EACH CATEGORY

SGSE	GPOD	$AP_{50:95}$	AP_{50}	AP_{75}	A220	A320	A330	ARJ21	B737	B787	Other
RT-DETR		45.22	62.67	49.67	58.84	80.09	85.32	62.37	33.96	57.93	60.18
✓		48.48	65.72	52.53	64.75	85.43	85.47	64.47	38.73	55.40	65.79
	✓	49.03	65.65	52.71	64.27	90.02	84.84	66.38	37.68	56.18	60.17
✓	✓	51.35	69.47	56.44	70.09	94.81	93.14	67.39	43.45	53.09	64.34

TABLE V

ABLATION ANALYSIS ON SARDET-100K [6], WITH $AP_{50:95}$ REPORTED FOR EACH CATEGORY

SGSE	GPOD	$AP_{50:95}$	AP_{50}	AP_{75}	Ship	Aircraft	Car	Tank	Bridge	Harbor
RT-DETR		60.38	90.33	63.86	72.07	48.10	66.76	52.23	47.42	76.18
✓		61.02	90.69	64.49	72.20	45.58	66.22	53.56	49.15	79.91
	✓	60.76	90.55	64.04	72.79	47.01	66.34	54.43	47.26	77.39
✓	✓	61.53	90.88	64.65	72.33	45.48	66.47	54.51	49.78	80.75

TABLE VI

INFLUENCE OF DIFFERENT ATTENTION MECHANISMS IN SGSE ON PERFORMANCE AND EFFICIENCY

Attention Type	$AP_{50:95}$	AP_{50}	AP_{75}	FLOPs
Self Attention	49.72	67.76	54.46	48.96G
Deformable Attention	51.35	69.47	56.44	47.02G

weakens target discrimination. The optimal result is obtained at $\rho = 50$.

To examine the spatial distributions learned by HSN across feature scales, Fig. 10 visualizes the predicted saliency maps for sparse targets, densely distributed tanks, large harbor regions, and an elongated bridge. At the deep level $l = 3$, the responses are relatively smooth and spatially coherent, providing coarse target-centered semantic localization while suppressing large background areas. At the intermediate level $l = 2$, the responses become more compact, better separate adjacent targets, and begin to recover target outlines and

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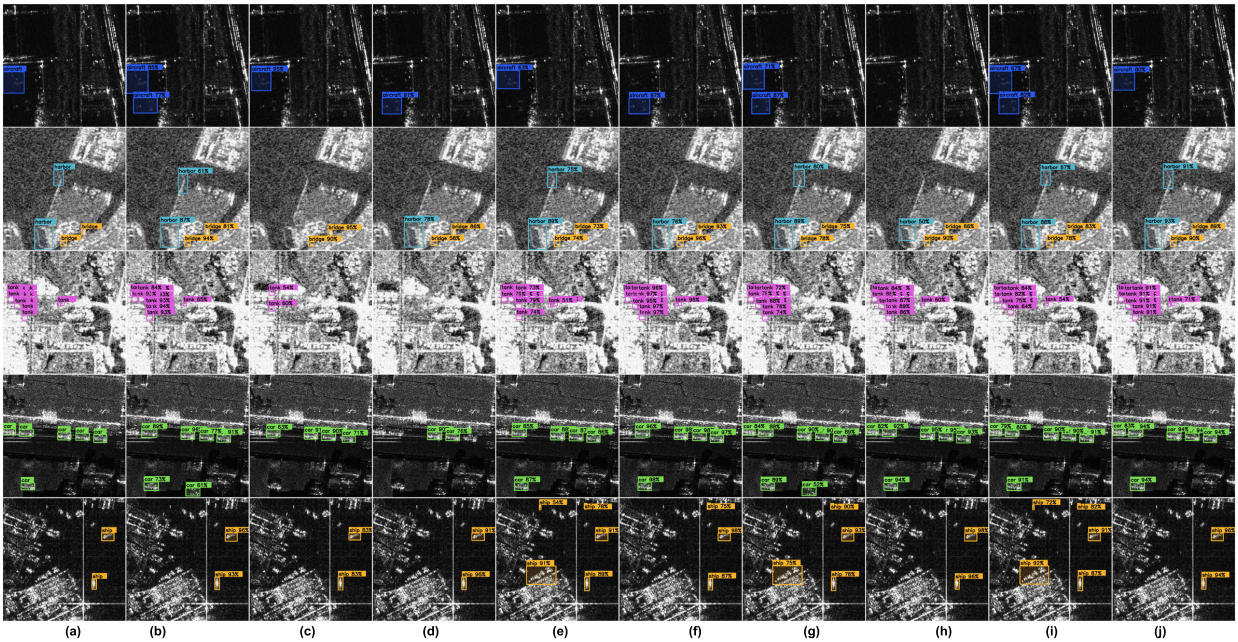


Fig. 7. Detection visualization on SARDet-100K [6] for representative end-to-end methods. Correct predictions are shown with green boxes, false alarms with red boxes, and missed objects with yellow circles. All results are obtained using a confidence threshold of 0.5. (a) GT. (b) Deformable DETR. (c) DAB-DETR. (d) Conditional DETR. (e) RT-DETR. (f) DINO. (g) DFINE. (h) DEIM. (i) RF-DETR. (j) SP-DETR (ours).

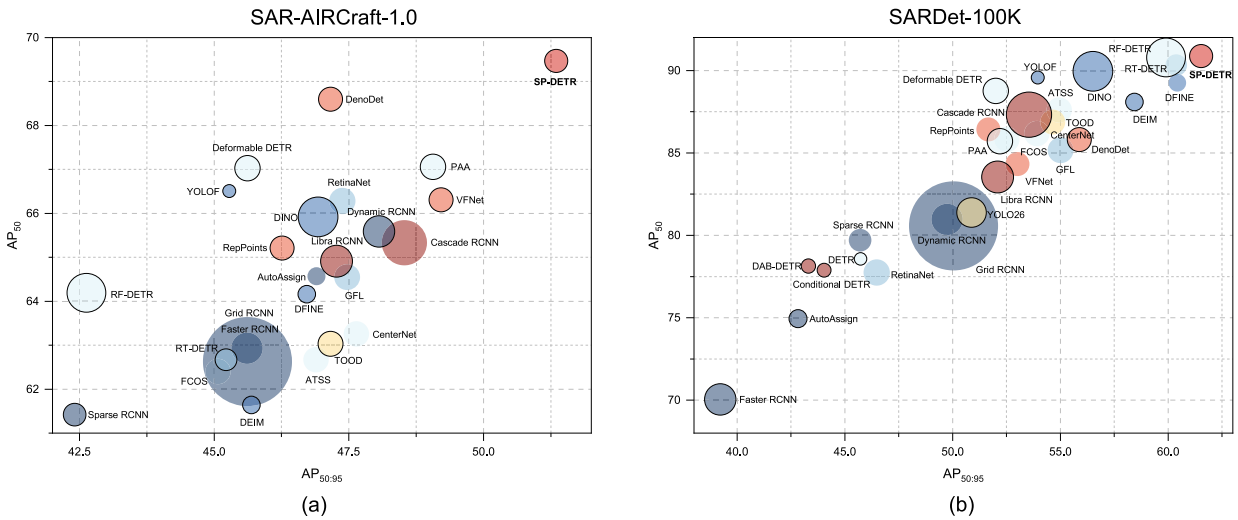


Fig. 8. Bubble charts of performance for different methods on the SAR-AIRCraft-1.0 and SARDet-100K datasets. The x- and y-axes represent the AP50:95 and AP50 of the models, respectively, while the bubble size denotes the required FLOPs. It can be observed that the proposed SP-DETR achieves precise detection of SAR targets with relatively low computational complexity. (a) Bubble chart for different methods on the SAR-AIRCraft-1.0 dataset. (b) Bubble chart for different methods on the SARDet-100K dataset.

internal scattering variations. At the shallow level $l = 1$, the maps exhibit more localized and discontinuous activations that preserve fine scattering structures, including the separated responses of dense tanks and the elongated structure of the bridge. Some isolated background activations also become more visible at this level, which is consistent with shallow SAR features containing both detailed target cues and clutter responses. These scale-dependent distributions support the intended coarse-to-fine recursion: aligned deep saliency biases residual modulation toward semantically relevant neighborhoods, while the local–global predictor at each shallower resolution restores scale-specific scattering details instead of copying the deep saliency map.

2) Analysis of GPOD: GPOD treats the predicted Gaussian box distributions as a stochastic localization policy and jointly optimizes them through supervised and reinforcement learning. To isolate the effect of reward granularity, Table VIII compares supervised learning alone with instance- and image-level reward variants under the same experimental setting. Compared with the supervised-learning baseline, the instance-level reward improves $AP_{50:95}$, AP_{50}, and AP_{75} by 1.70, 3.07, and 0.59 points, respectively, whereas the image-level reward yields improvements of 2.87, 3.75, and 3.91 points. More importantly, the image-level reward outperforms its instance-level counterpart by 1.17 $AP_{50:95}$, 0.68 AP_{50}, and 3.32 AP_{75}. The substantially larger improvement at AP_{75} is consistent

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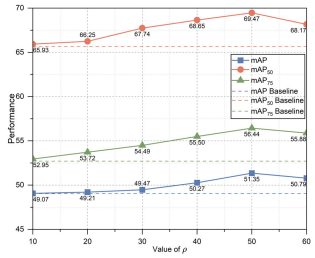


Fig. 9. Impact of varying ρ in SGSE on $mAP_{50:95}$, mAP_{50}, and mAP_{75} on the SAR-AIRCraft-1.0 dataset, as well as comparisons with corresponding baselines. The dashed lines in the figure denote the corresponding baselines.

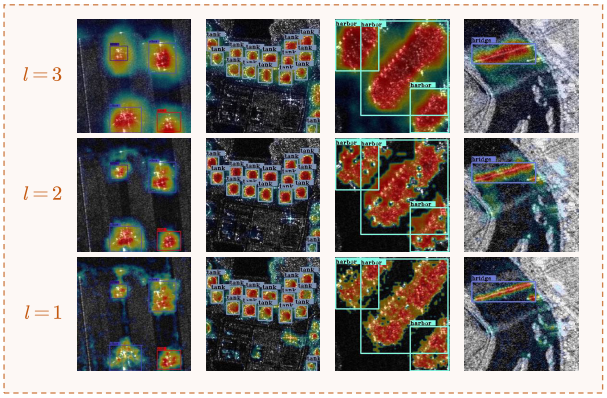


Fig. 10. Spatial distributions of the HSN saliency maps at three feature levels when $\rho = 50$. From the deep level $l = 3$ to the shallow level $l = 1$, the maps transition from coarse target-centered semantic responses to increasingly localized scattering structures.

TABLE VII
COMPONENTWISE ABLATION OF SGSE ON SAR-AIRCRAFT-1.0

Variant	SGSE Components			Detection Performance		
	CLG	SI	SAMM	$AP_{50:95}$	AP_{50}	AP_{75}
Aux. Sup.	✓			49.57	67.42	52.35
w/o SAMM	✓	✓		49.69	67.94	53.88
w/o CLG		✓	✓	51.01	68.87	54.30
Full	✓	✓	✓	**51.35**	**69.47**	**56.44**

TABLE VIII
IMPACT OF DIFFERENT OPTIMIZATION METHODS ON PERFORMANCE

Method	$AP_{50:95}$	AP_{50}	AP_{75}
SL-only	48.48	65.72	52.53
Instance-level Reward	50.18	68.79	53.12
Image-level Reward	51.35	69.47	56.44

with the intended benefit for stricter localization. This difference reflects the complementary optimization granularities of the two branches. The supervised branch provides instancewise box-regression gradients, while the image-level reward applies a common advantage to the joint localization policy of all matched predictions. The reinforcement learning objective, therefore, complements the supervised objective by evaluating matched predictions jointly at the image level.

Table IX further evaluates the reward function and assignment strategy under the complete SP-DETR setting, where CBM and IBM denote CBM and independent bipartite

TABLE IX
ABLATION OF THE GPOD REWARD AND ASSIGNMENT ON SAR-AIRCRAFT-1.0

Assignment	IoU Reward	Detection Performance		
		$AP_{50:95}$	AP_{50}	AP_{75}
IBM	GIoU	50.22	67.69	55.46
CBM	IoU	51.27	69.05	55.88
CBM	DIoU	51.08	69.44	55.30
CBM	CIoU	51.30	69.12	55.64
CBM	GIoU	**51.35**	**69.47**	**56.44**

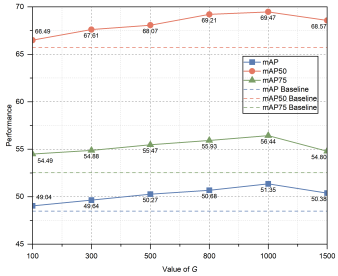


Fig. 11. Impact of varying G in GPOD on $mAP_{50:95}$, mAP_{50}, and mAP_{75} using SAR-AIRCraft-1.0, together with corresponding baselines. The dashed lines in the figure denote the corresponding baselines.

matching, respectively. The GIoU entries in the table use the normalized GIoU reward defined above. The last four rows retain CBM and vary only the reward function. GIoU achieves 51.35 $AP_{50:95}$, 69.47 AP_{50}, and 56.44 AP_{75}. A likely reason for its better overall performance is its consistency with the supervised optimization objective. The standard RT-DETR box-regression loss contains a GIoU term, while the policy branch evaluates sampled boxes using normalized GIoU. The two branches, therefore, encourage compatible improvements in localization quality. The $AP_{50:95}$ advantage of GIoU over IoU, DIoU, and CIoU is modest, while the corresponding AP_{75} gains are 0.56, 1.14, and 0.80 points, respectively. These results support GIoU as the best overall choice among the evaluated rewards, particularly under the stricter overlap criterion.

The assignment-strategy comparison retains normalized GIoU and changes only the matching procedure. Compared with CBM, IBM decreases $AP_{50:95}$, AP_{50}, and AP_{75} by 1.13, 1.78, and 0.98 points, respectively. The result supports preserving a consistent query–target correspondence across stochastic samples.

To examine how the hyperparameter G in GPOD affects detection performance, we use RT-DETR equipped with SGSE, where ρ is fixed at 50 as the baseline. We evaluate six values of G, i.e., 10, 100, 300, 500, 1000, and 1500, on SAR-AIRCraft-1.0 and report the results in Fig. 11. In GPOD, G specifies how many policy samples are drawn, thereby controlling the strength of reinforcement learning optimization during training. The results show that increasing G leads to continuous improvements in $AP_{50:95}$, AP_{50}, and AP_{75}. When G reaches 1000, the model performance achieves its optimum, but, when G further increases to 1500, the performance declines. These results indicate that during model training, it

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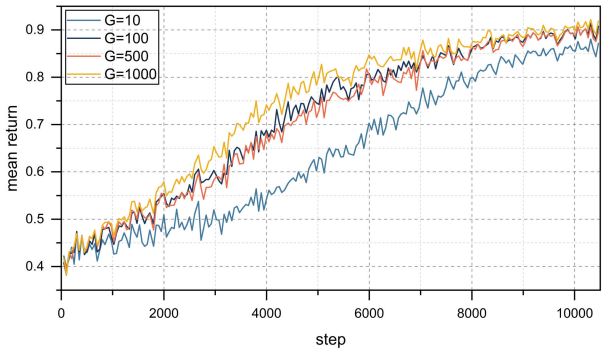


Fig. 12. Variation curves of the average reward for the last decoder layer during the model training process when G takes values of 10, 100, 500, and 1000.

is necessary to properly balance the two optimization mechanisms of supervised learning and reinforcement learning. An excessively large or small G may break this optimization equilibrium, leading to a performance drop. Consequently, G is set to 1000 based on these results. In addition, we further plot the variation curves of the average reward for the last decoder layer during the training process when G takes values of 10, 100, 500, and 1000. As shown in Fig. 12, within a certain range, a larger G is beneficial for training the policy model in the decoder. Increasing G causes only a modest increase in training-time computation and GPU memory because it expands the low-dimensional box-level tensors used for sampling, log-probability, and reward evaluation, while the backbone, encoder, and decoder feature computations are shared across all groups. Because group sampling and reward evaluation are disabled during inference, G does not affect inference memory consumption or latency.

V. Conclusion

This work presents SP-DETR, an end-to-end detector for SAR object detection. The SGSE module uses multiscale saliency estimates to select high-value shallow tokens for sparse interaction and reweight fused features, improving the utilization of local target details while suppressing background interference. The GPOD module models each predicted bounding box as a Gaussian distribution and combines multigroup sampling with a global localization reward to improve localization robustness under ambiguous target extents. Experiments on SAR-AIRCraft-1.0 and SARDet-100K show that SP-DETR achieves the best overall $AP_{50:95}$ among the compared methods.

Future work will investigate SAR object detection under more challenging conditions, focusing on robustness and generalization in scenes with dense target distributions, small objects, and severe clutter interference. Furthermore, we aim to extend SP-DETR to low-annotation-cost scenarios, including few-shot and semisupervised learning, thereby improving its applicability in practical deployments.

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